

**PAPER
B**



2020 SCIENCE

TIME ALLOWED: 45 MINUTES

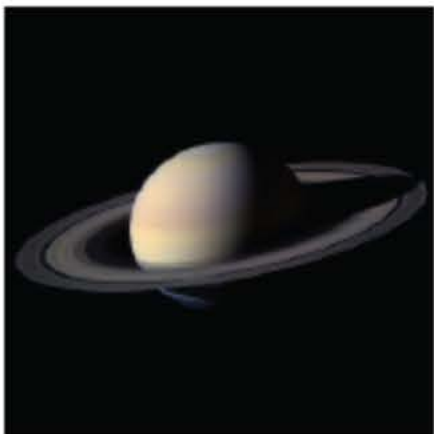
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Paula's teacher gave her four photos, each one showing a different object in space.

She also told Paula four facts. Each fact was about one of these objects.

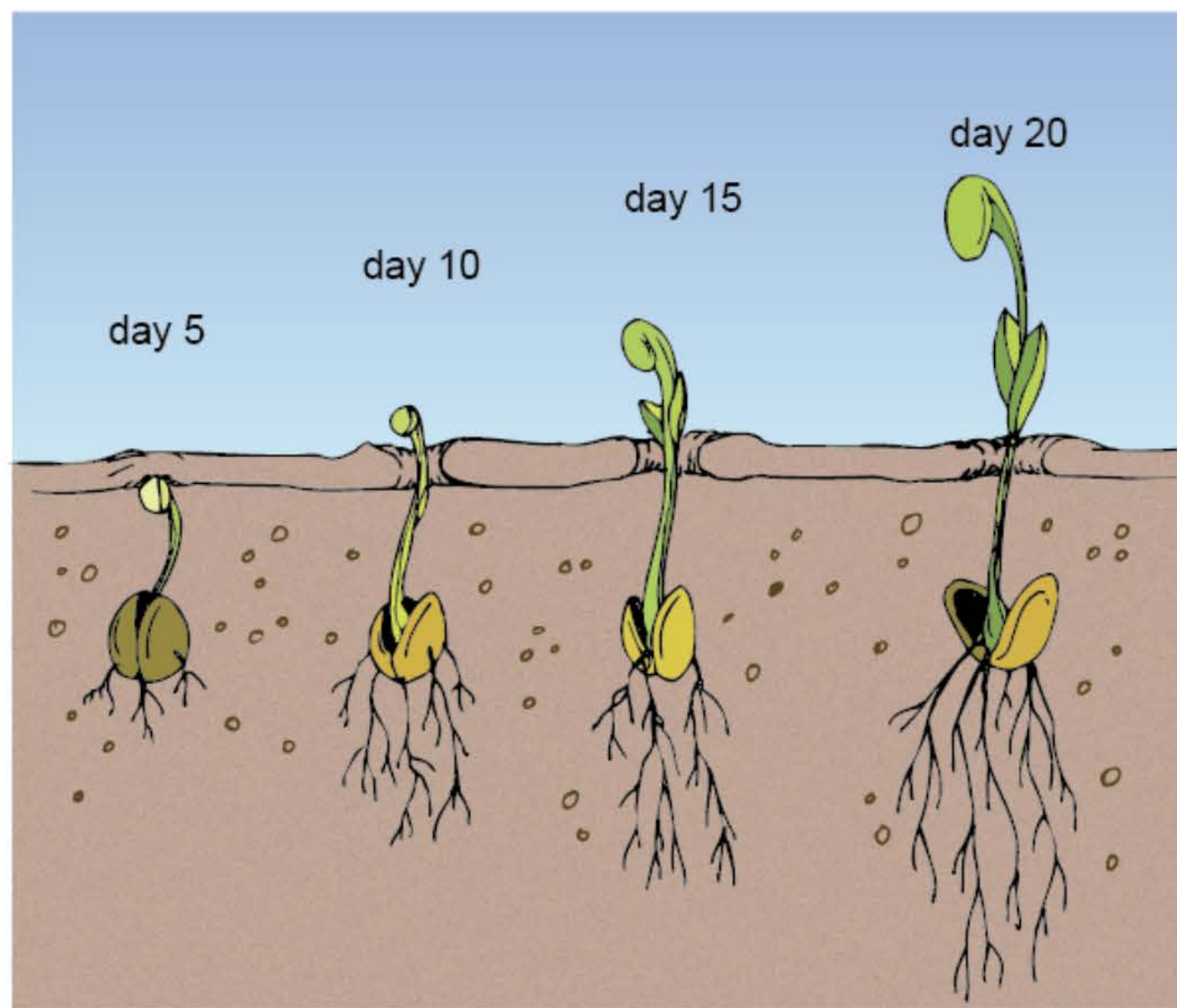
- Saturn is a planet with rings around it.
- A comet is a ball of frozen gases, leaving a trail behind it.
- An asteroid is a lump of rock that orbits the Sun.
- Mars is a red planet.

Which photo shows an asteroid?

☐☐☐☐

Photos courtesy of NASA/JPL-Caltech/Mattias/MSFC/Aaron Kingery.

The diagram shows a growing plant on different days.



When does the plant first grow above the ground?

☐

day 5

☐

between day 5 and day 10

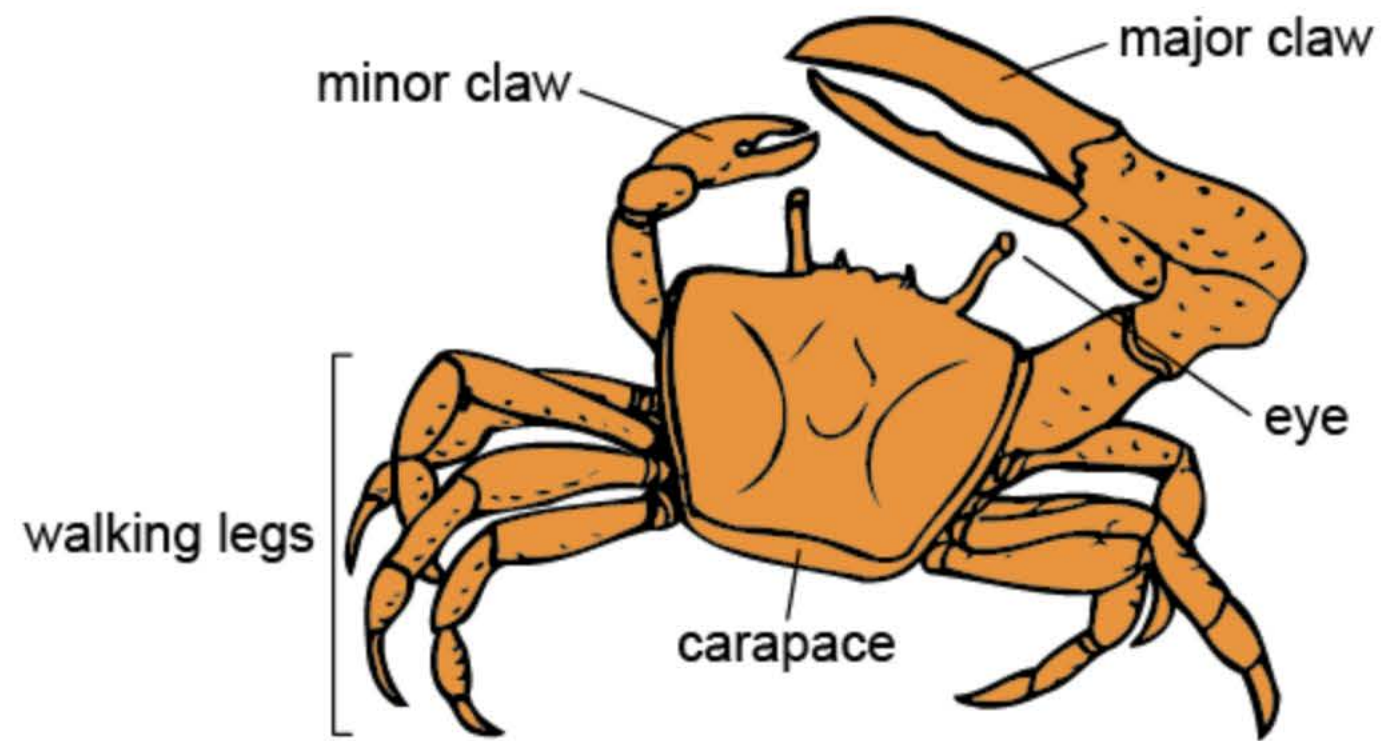
☐

day 10

☐

between day 10 and day 15

This is a male fiddler crab.



How many pairs of walking legs does it have?

☐

10

☐

5

☐

2

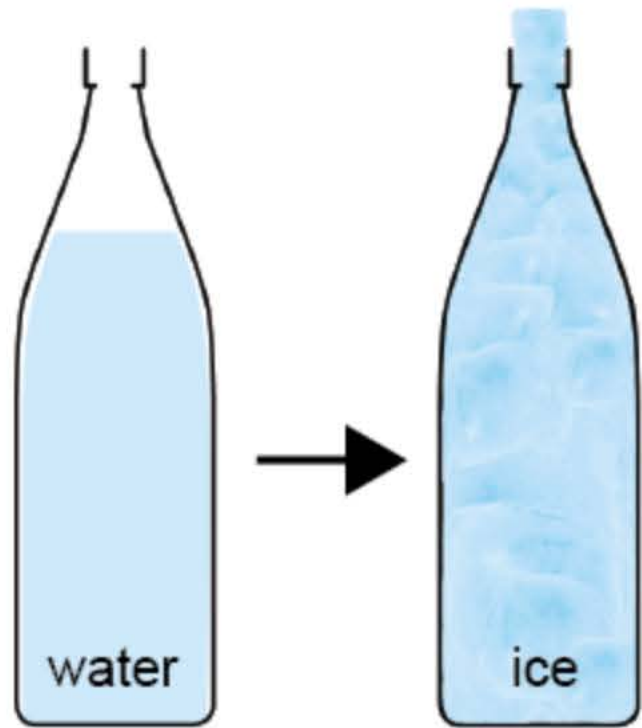
☐

8

☐

4

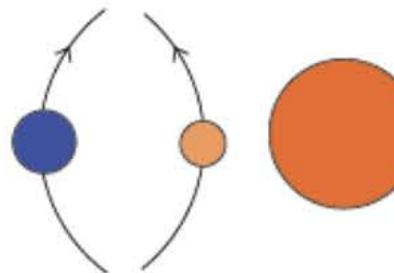
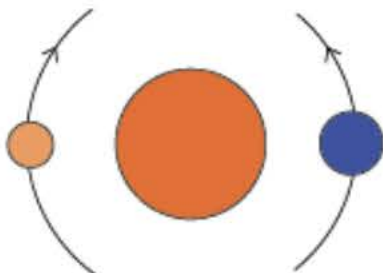
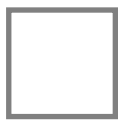
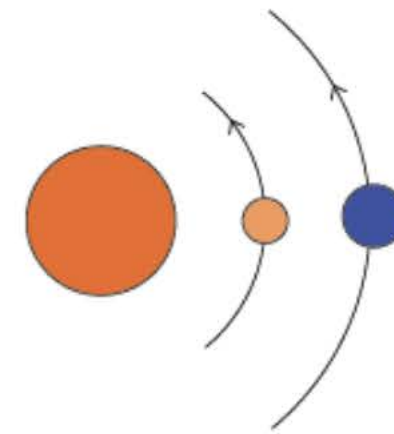
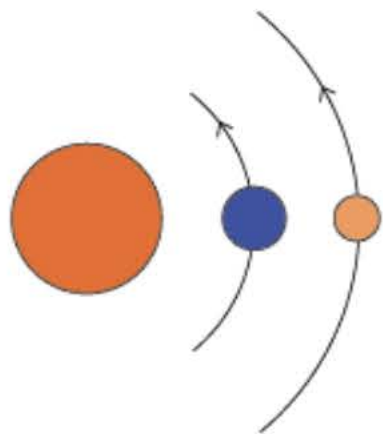
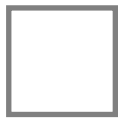
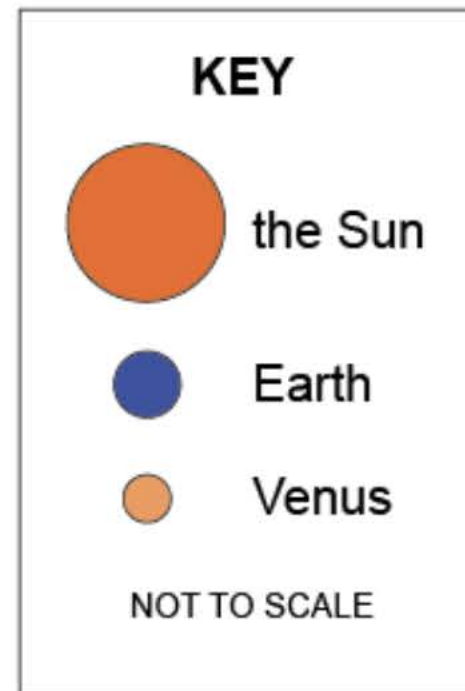
Jason placed a bottle of water in the freezer. When he took the bottle of water out of the freezer the next morning, this is what he saw.



What is the best explanation for this?

- ☐ Water takes up more space when frozen.
- ☐ Water becomes heavier when frozen.
- ☐ Water bubbles up when freezing.
- ☐ Water spills before freezing.

The transit of Venus occurs when the planet Venus travels between the Sun and Earth. It can be observed from Earth. Which diagram correctly shows the position of the Sun, Venus and Earth when a transit of Venus takes place?



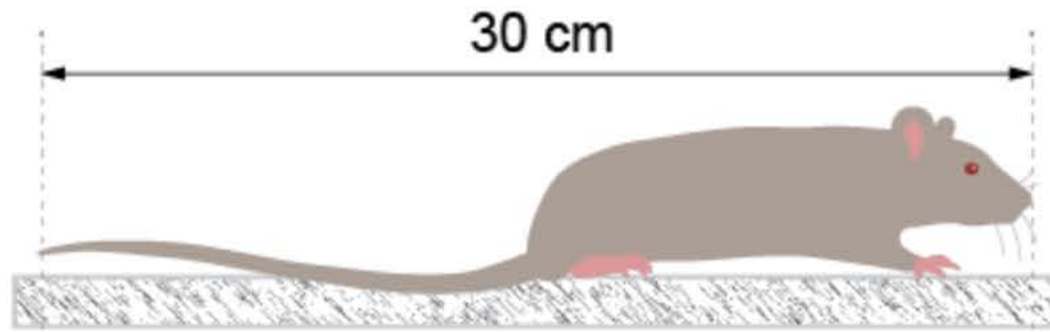
Solar panels convert energy from sunlight into electricity.



Which of the following would affect the amount of energy captured by the solar panels?

- ☐ how hot the day is
- ☐ how strong the roof is
- ☐ how much energy is needed
- ☐ how sunny it is during the day

This is a drawing of a rat.



About how many centimetres long is the rat's tail?

☐

5

☐

10

☐

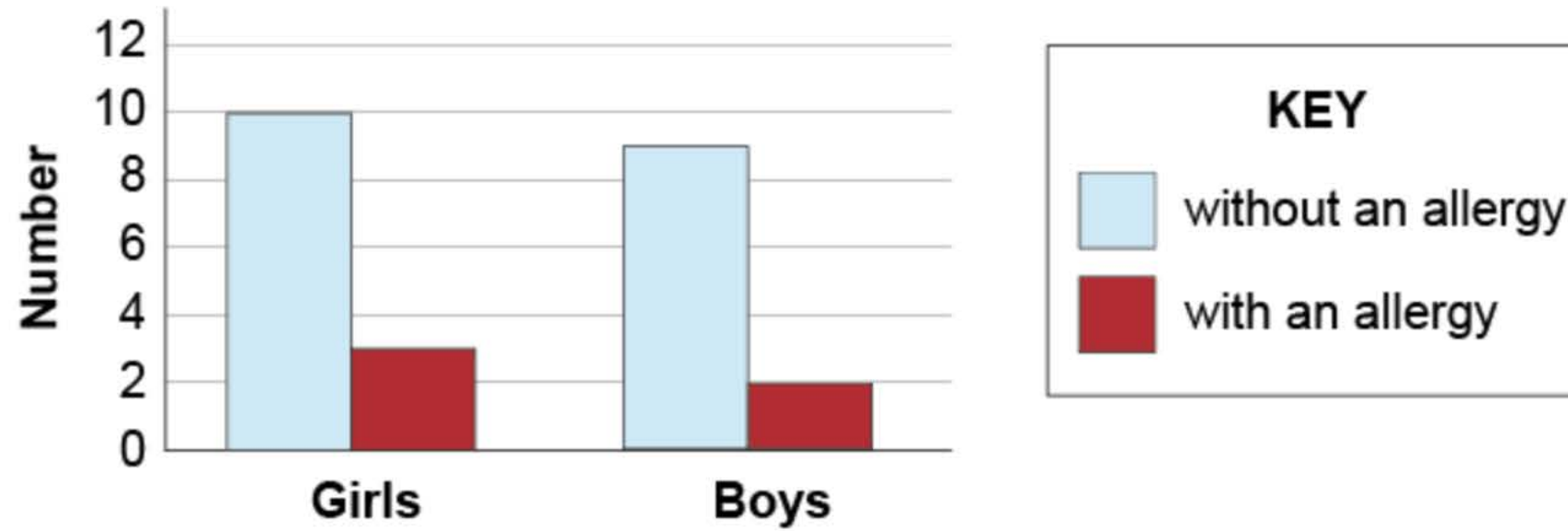
15

☐

20

The graph shows the results of a student allergy survey in one class.

Student Allergy Survey



Which of the following groups has the greatest number of students in it?

☐

students with an allergy

☐

girls with an allergy

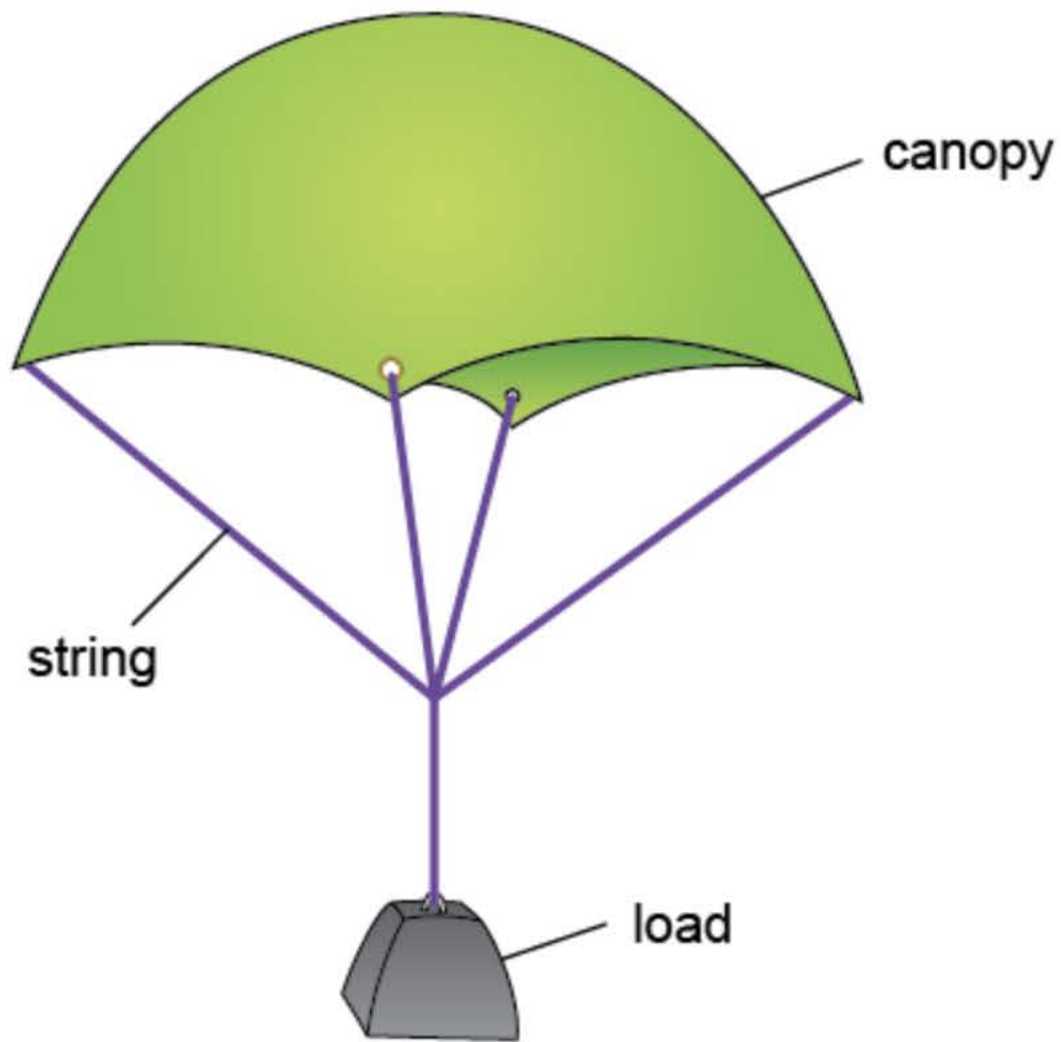
☐

boys without an allergy

☐

girls without an allergy

A simple parachute can be made of a canopy and some strings. Four students each made a parachute using different materials.



The students wanted to test which parachute would stay in the air the longest.

They took their parachutes to the second floor. They held their parachutes so that the load of each one was at the same height. Then they dropped their parachutes at the same time.

To make the test fair

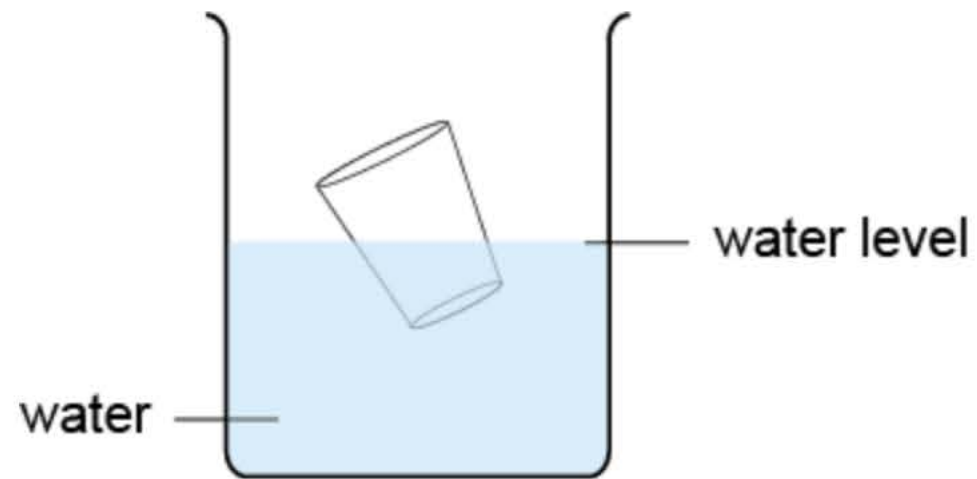
☐ the canopies should be made from the same material.

☐ the loads should be the same.

☐ the strings should be the same length.

☐ the parachutes should weigh the same.

Lisa placed an empty plastic cup in a container of water.
The diagram shows the plastic cup floating in the water.



What will happen if Lisa places a marble into the plastic cup?

- ☐ The water level will move downwards.
- ☐ The water level will move upwards.
- ☐ The water level will not change.
- ☐ The amount of water in the container will increase.

To conduct a fair test, the investigators must:

- **change** only one thing for each test,
- **keep** everything else the same, and
- **observe or measure** one change that happens.

A group of Year 3 students want to see if the shadow of a stick changes length during the day.

What must the students change, what must they keep the same, and what must they measure to make their investigation a fair test?

	Change	Keep the same	Measure
(A)	the time of the day	the length of the shadow	the length of the stick
(B)	the length of the shadow	the length of the stick	the time of the day
(C)	the time of the day	the length of the stick	the length of the shadow
(D)	the length of the stick	the time of the day	the length of the shadow

☐

A

☐

B

☐

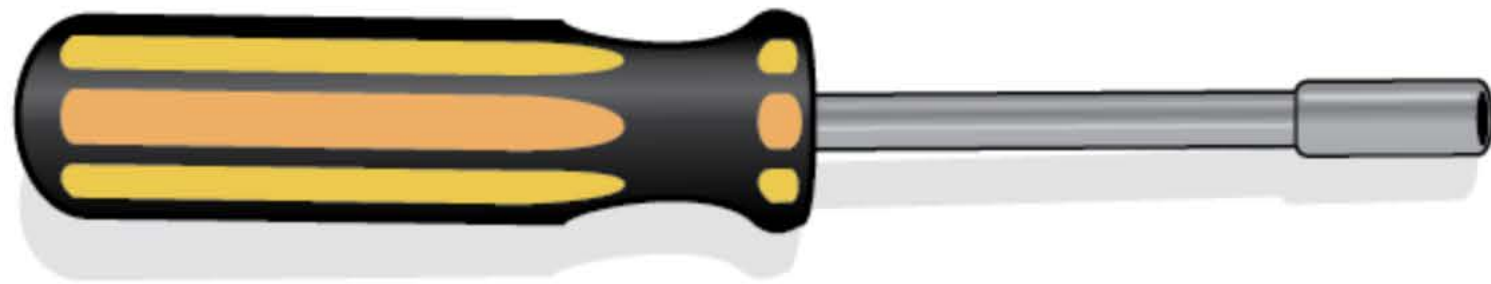
C

☐

D

A screwdriver turns a screw, which pushes the screw into wood. Different bits allow a screwdriver handle to be used with different screws.

screwdriver handle



bit set



screws



Each screw has a head with a particular shape. Only a bit with a matching shape will turn the screw.

These are some bits.



A



B



C



D

Which of the following screw heads matches bit C?

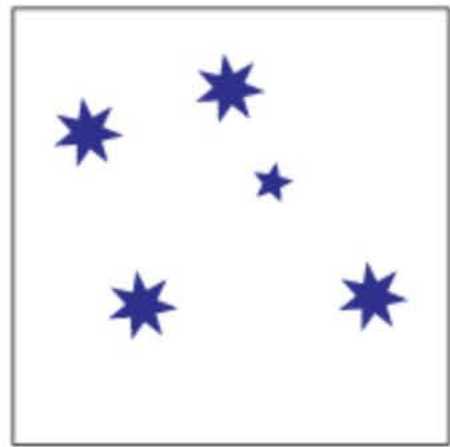
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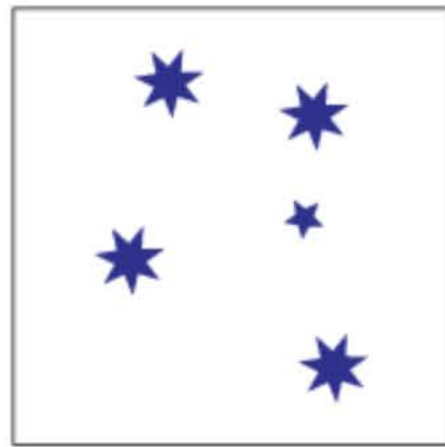
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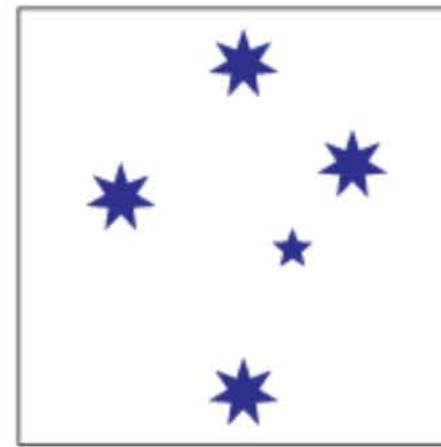

An astronomer observed a group of stars called the Southern Cross in the night sky. He drew the position of the stars every two hours, as shown in the diagram.



6 pm

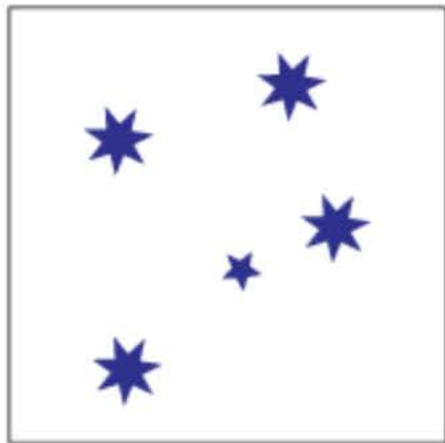
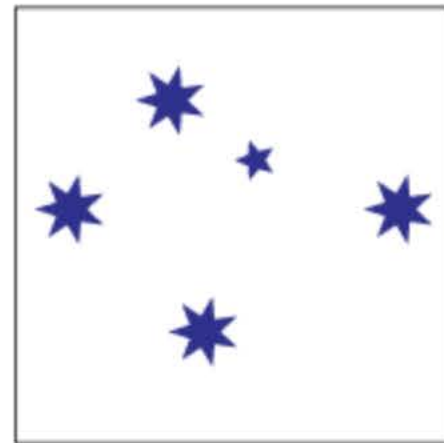
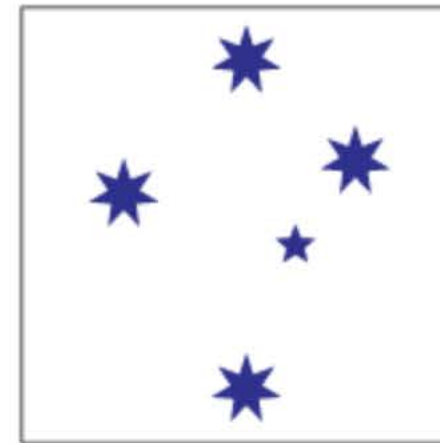
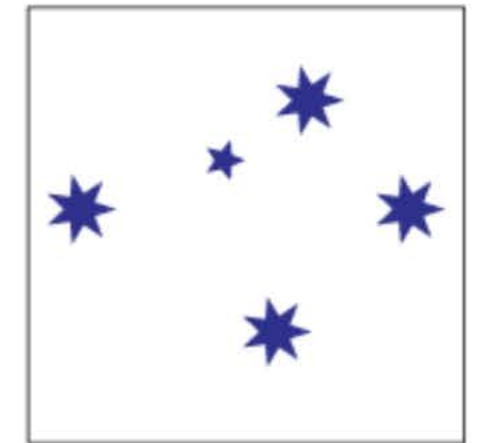


8 pm



10 pm

Which of these shows the position of the Southern Cross at midnight?

☐

☐

☐

☐


Brian is looking at the photographs of some animals. All the animals have antennae or pedipalps on their heads.



Halictus scabiosae image by Didier Descouens https://commons.wikimedia.org/wiki/File:Halictus_scabiosae_MHNT.jpg, CC BY-SA 4.0.

He places the animals into two groups, based on one particular feature.

Group 1: G, H, K

Group 2: I, J

What feature does he use to decide which group to place them in?

	Group 1	Group 2
(A)	spots	no spots
(B)	six legs	eight legs
(C)	a large body	a small body
(D)	wings	no wings

☐

A

☐

B

☐

C

☐

D

In the first photograph the arrow shows the length of a cicada's body.

In the second photograph the arrow shows the width of a cicada's thorax.



Body length



Thorax width

Which option has both the correct body length and correct thorax width?

- ☐ Body length is $4\frac{1}{2}$ cm Thorax width is $2\frac{1}{2}$ cm
- ☐ Body length is $4\frac{1}{2}$ cm Thorax width is 2 cm
- ☐ Body length is $6\frac{1}{2}$ cm Thorax width is $2\frac{1}{2}$ cm
- ☐ Body length is $6\frac{1}{2}$ cm Thorax width is 2 cm

Three students observed popcorn being made.



Kim observed: 'There were lots of little popping sounds and a smell like cooked butter.'

Peter observed: 'It was yellow in colour and had a round shape.'

Noor observed: 'It was hot in my hand and wasn't very heavy.'

Which statement about the observations is correct?

- ☐ Kim used her sense of smell and her sense of hearing.
- ☐ Peter used his sense of sight and his sense of taste.
- ☐ Kim and Noor both used their sense of smell.
- ☐ All three students used their sense of sight.

A key allows users to identify an object, such as an animal or a plant, using the characteristics of the object.

The diagram shows a key for a group of animals called vertebrates.

1.	has feathers	bird
	does not have feathers	go to 2
2.	has scales	go to 3
	does not have scales	go to 4
3.	has gills	fish
	does not have gills	reptile
4.	has fur	mammal
	does not have fur	amphibian

A caecilian does not have feathers, does not have scales and does not have fur.

What type of vertebrate is a caecilian?

☐

fish

☐

reptile

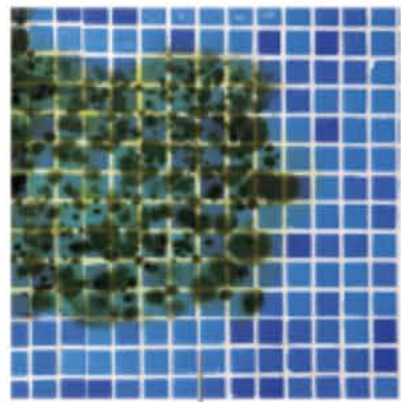
☐

mammal

☐

amphibian

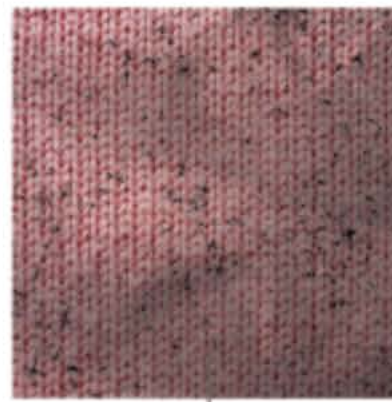
Angela wanted to test how well four different mould removal sprays worked. She sprayed each product on a mouldy surface in her home.

Product 1

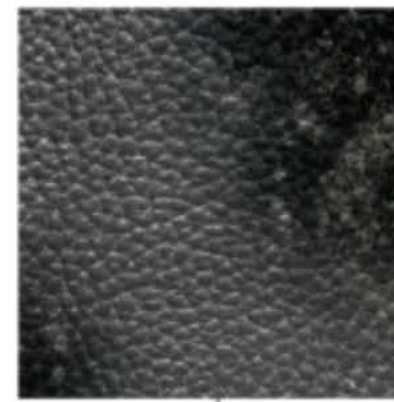
bathroom floor

Product 2

sink

Product 3

fabric

Product 4

leather

Angela left each spray in place for ten minutes before wiping the surfaces and comparing the results.

Which of these options gives correct information about the investigation?

	Is this a fair test?	Why?
(A)	Yes	because different surfaces were tested
(B)	No	because 10 minutes was not long enough
(C)	Yes	because all methods to clean the surfaces were the same
(D)	No	because the products were not tested on the same surface

☐

A

☐

B

☐

C

☐

D

Soil erosion is the wearing away of the upper layer of soil.

Rainfall run-off is the rainwater that is not absorbed by the soil but flows across the soil surface.

The table shows the results of an investigation into the effect of vegetation cover on soil erosion and rainfall run-off after a heavy storm.

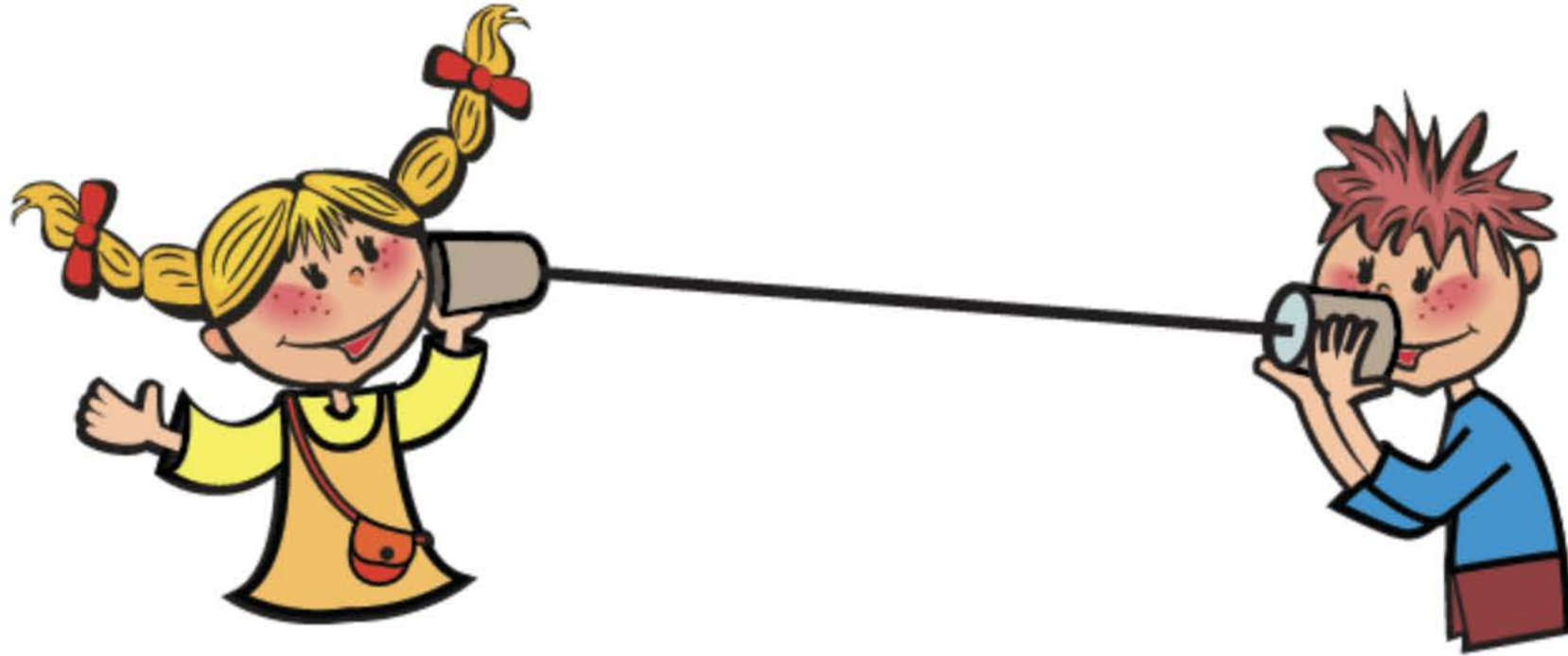
Factor	Treatment area		
	A	B	C
Area of soil covered by vegetation (%)	87	69	6
Amount of rainfall run-off (mm)	3	26	70
Amount of soil lost (tonnes/hectare)	0.03	0.3	22

Which statement best summarises the results?

- ☐ High percentage of vegetation cover results in high amount of rainfall run-off.
- ☐ High amount of rainfall run-off results in low amount of soil lost.
- ☐ Low percentage of vegetation cover results in low amount of soil lost.
- ☐ Low percentage of vegetation cover results in high amount of soil lost.

Sound can travel through different materials. When we speak to each other, the sound travels through the air to reach our ears.

Jill and David want to find out whether the material that a sound travels through has an effect on how it sounds. They made a string telephone like this one, using a cotton string and two cups.

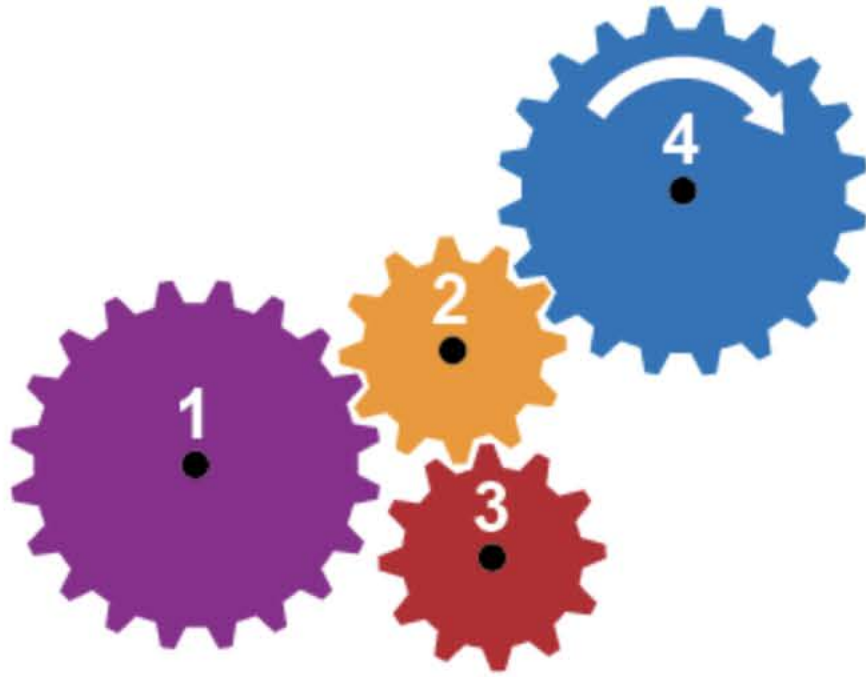


What else do they need for their investigation?

- ☐ a real telephone
- ☐ strings made of other materials
- ☐ cotton strings of different lengths
- ☐ other people to listen to the telephone

A gear is a wheel with teeth around its edge. When the teeth on the gears fit together, one gear can make another gear turn.

Alice put together a set of gears, as shown.



Which statement is correct?

☐

Gears 1, 2 and 3 turn in the same direction.

☐

Gear 1 turns in the same direction as gear 3.

☐

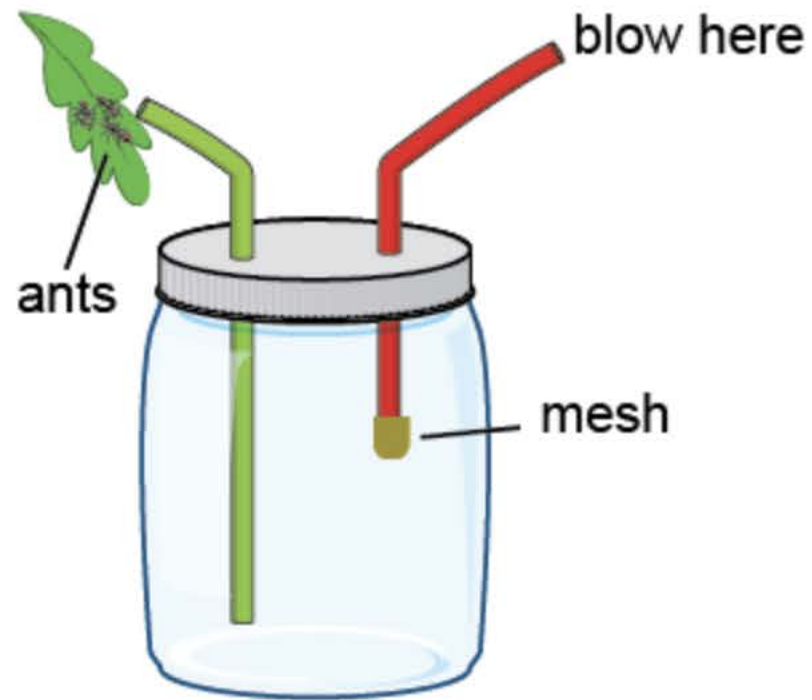
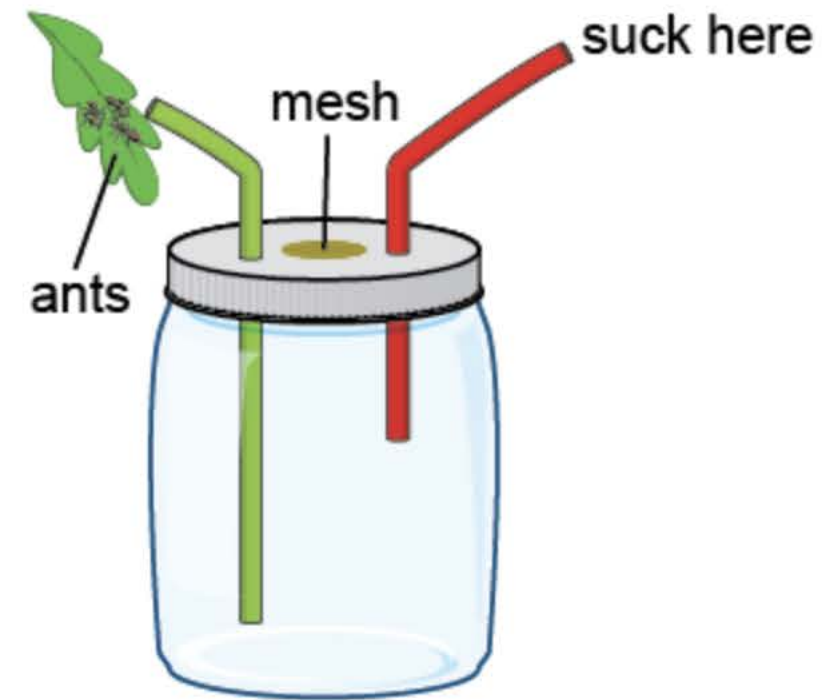
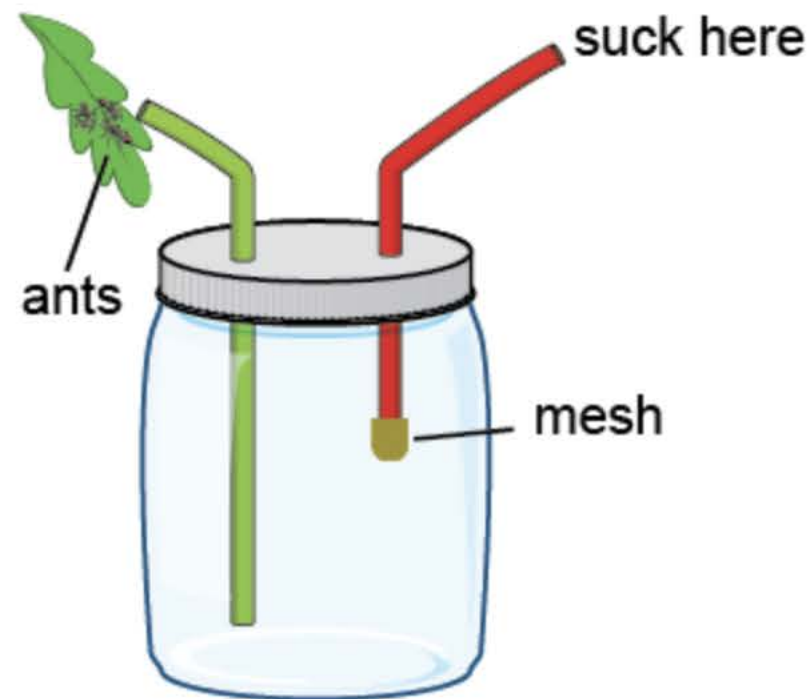
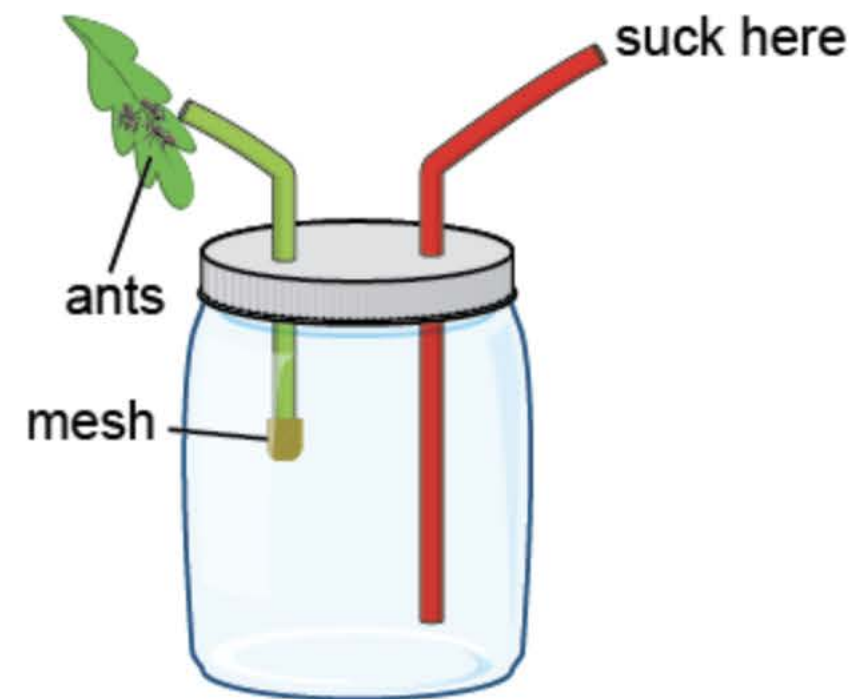
Gear 3 turns in the opposite direction to gear 4.

☐

Gear 4 turns in the opposite direction to gear 1.

Priya wanted to make a pooter to collect insects in the school garden. She used a glass jar with holes cut in the lid, two straws and some fine mesh.

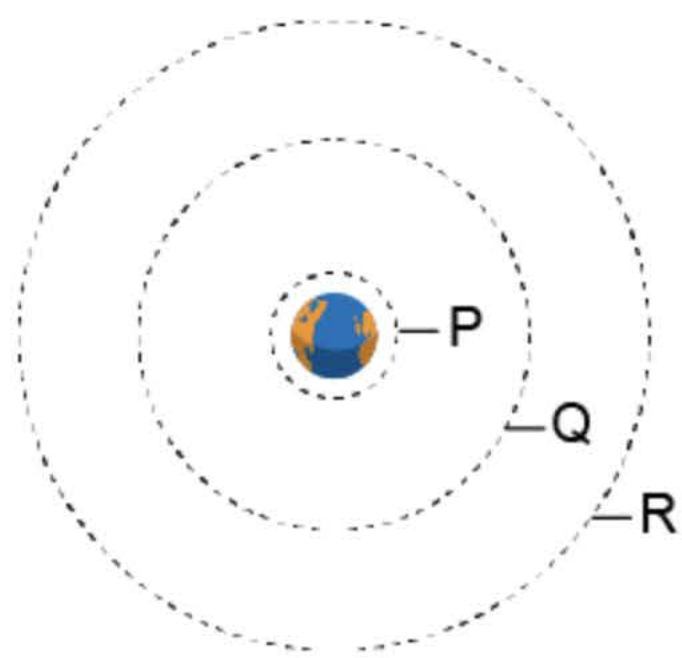
How should Priya make and use her pooter in order to collect insects safely?

☐

☐

☐

☐


An artificial satellite is an object that has been launched into orbit around Earth by rockets. Artificial satellites vary in size, altitude and purpose.

Satellite(s)	Altitude (kilometres above Earth)
International communication satellites	36 000
Navigation satellites (GPS)	20 000
Hubble space telescope	200 to 2000

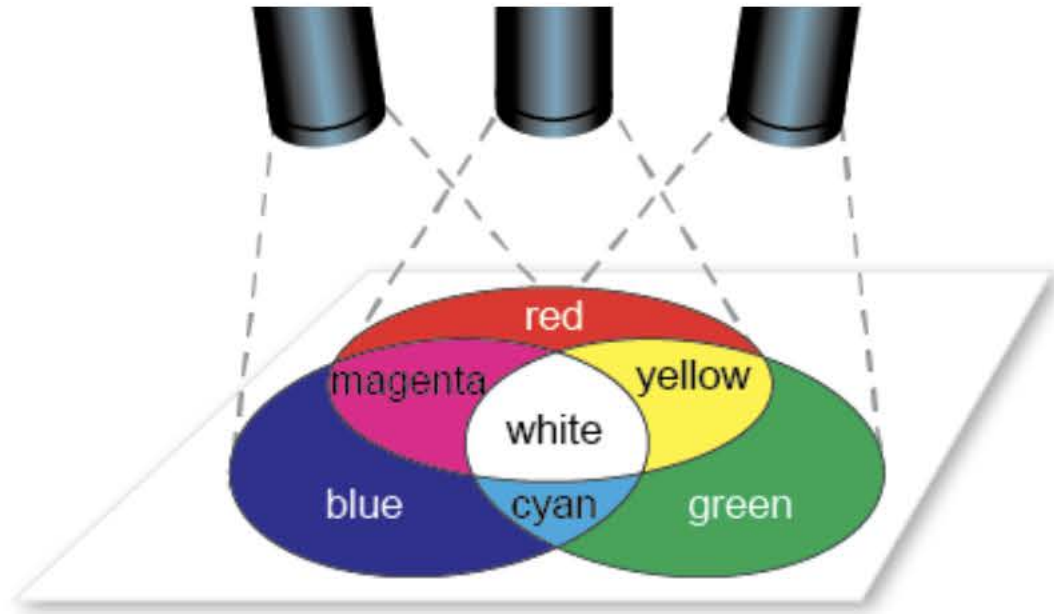
This scale diagram shows the three main orbits of satellites around Earth.



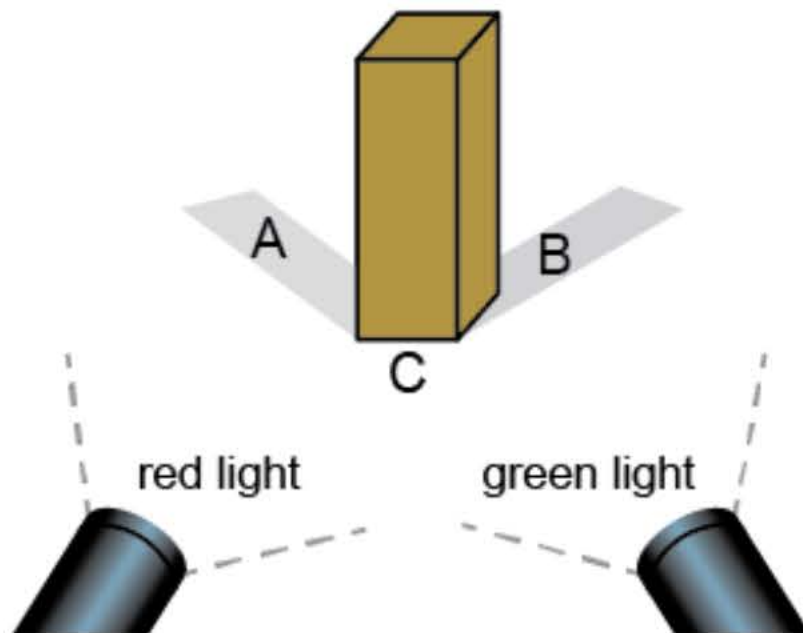
Which of these correctly match the satellite(s) with the orbits in the diagram?

- ☐ P: Hubble space telescope
Q: International communication satellites
R: Navigation satellites
- ☐ P: International communication satellites
Q: Hubble space telescope
R: Navigation satellites
- ☐ P: Navigation satellites
Q: International communication satellites
R: Hubble space telescope
- ☐ P: Hubble space telescope
Q: Navigation satellites
R: International communication satellites

Kate had three torches that each produced a light of a different colour. Kate conducted an investigation in a darkened room. She shone the torches onto a flat white surface so their light beams overlapped.



Then she placed a wooden block onto the white surface. She shone two torches at the block as shown in the diagram.



Which of the following correctly identifies the colour that areas A, B and C would appear?

☐

area A: red
area B: green
area C: red

☐

area A: red
area B: green
area C: yellow

☐

area A: red
area B: green
area C: white

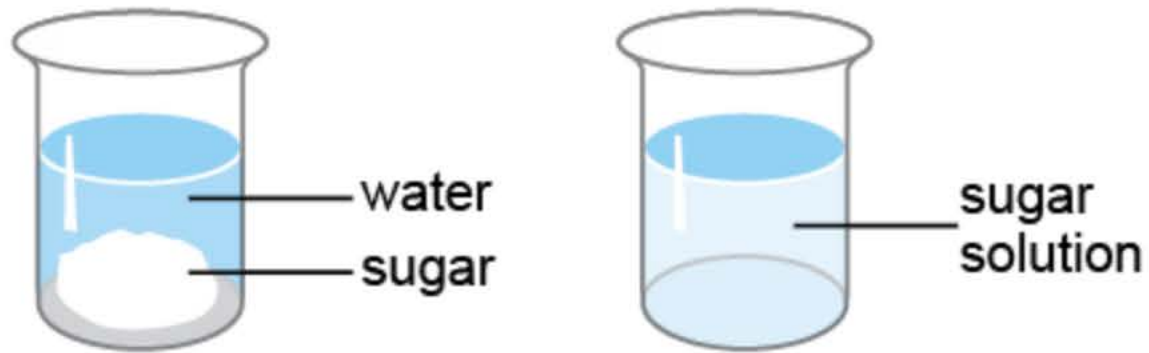
☐

area A: blue
area B: green
area C: cyan

☐

area A: white
area B: magenta
area C: yellow

When you add some sugar to a glass of water and stir for a while, the sugar crystals slowly disappear, leaving a clear liquid. We say the sugar dissolves in water, and the two together make a sugar solution.



The table shows how much sugar can dissolve in 100 grams of water at different temperatures.

Temperature (°C)	10	20	40	60
Amount of sugar (grams)	180	200	240	280

How many grams of sugar can dissolve in 50 grams of water at 20 °C?

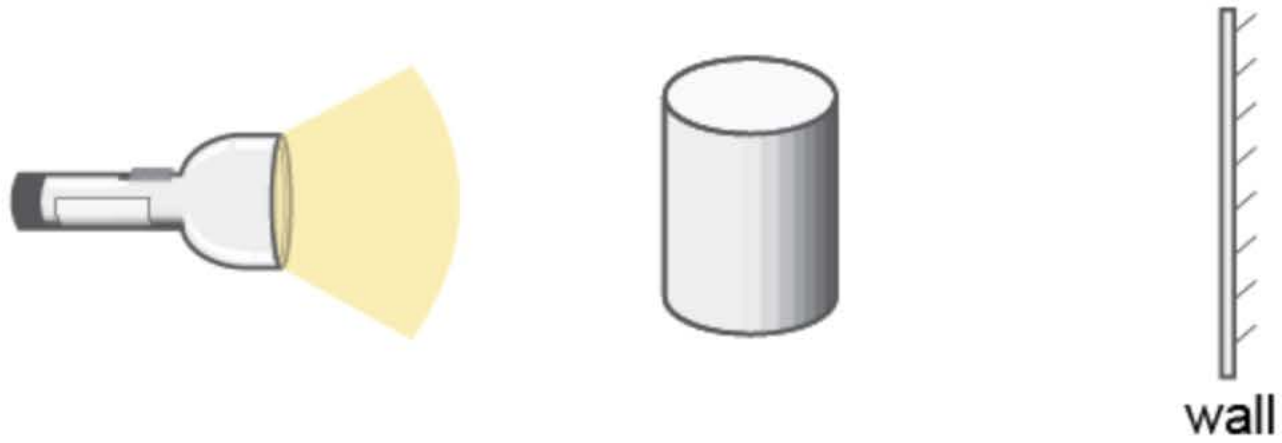
100

150

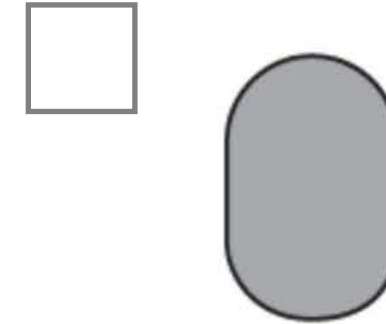
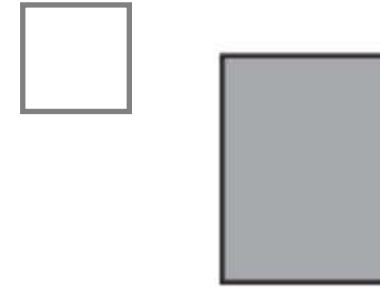
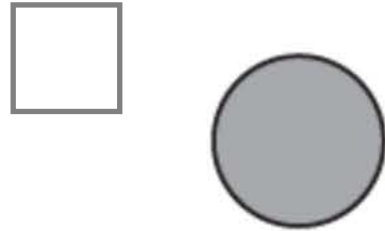
200

250

Joyce shines a torch onto a cylinder, as shown.

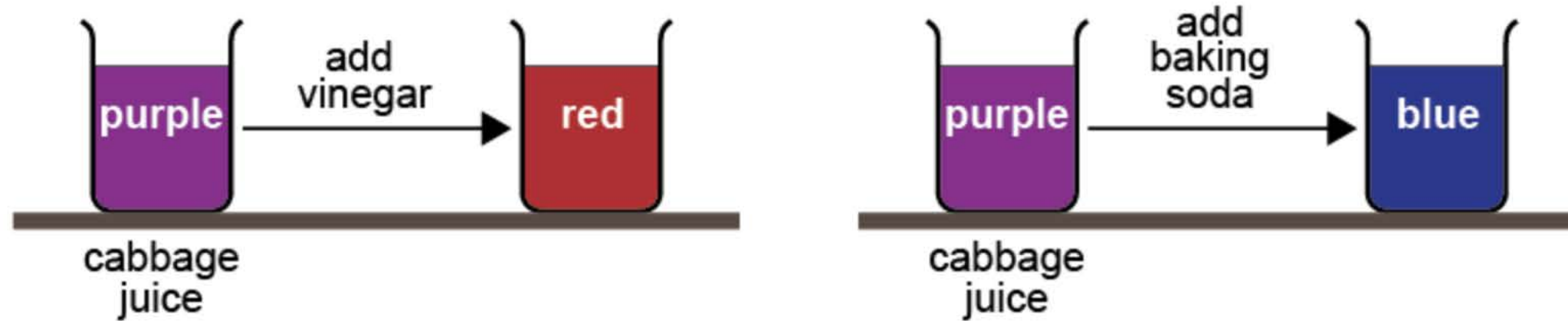


Which diagram shows the shadow produced by the cylinder on the wall?



Purple cabbage juice can be used to indicate whether a substance is alkaline or acidic.

Max took two cups of purple cabbage juice. He then added vinegar to one and baking soda to the other and noted the colour changes.

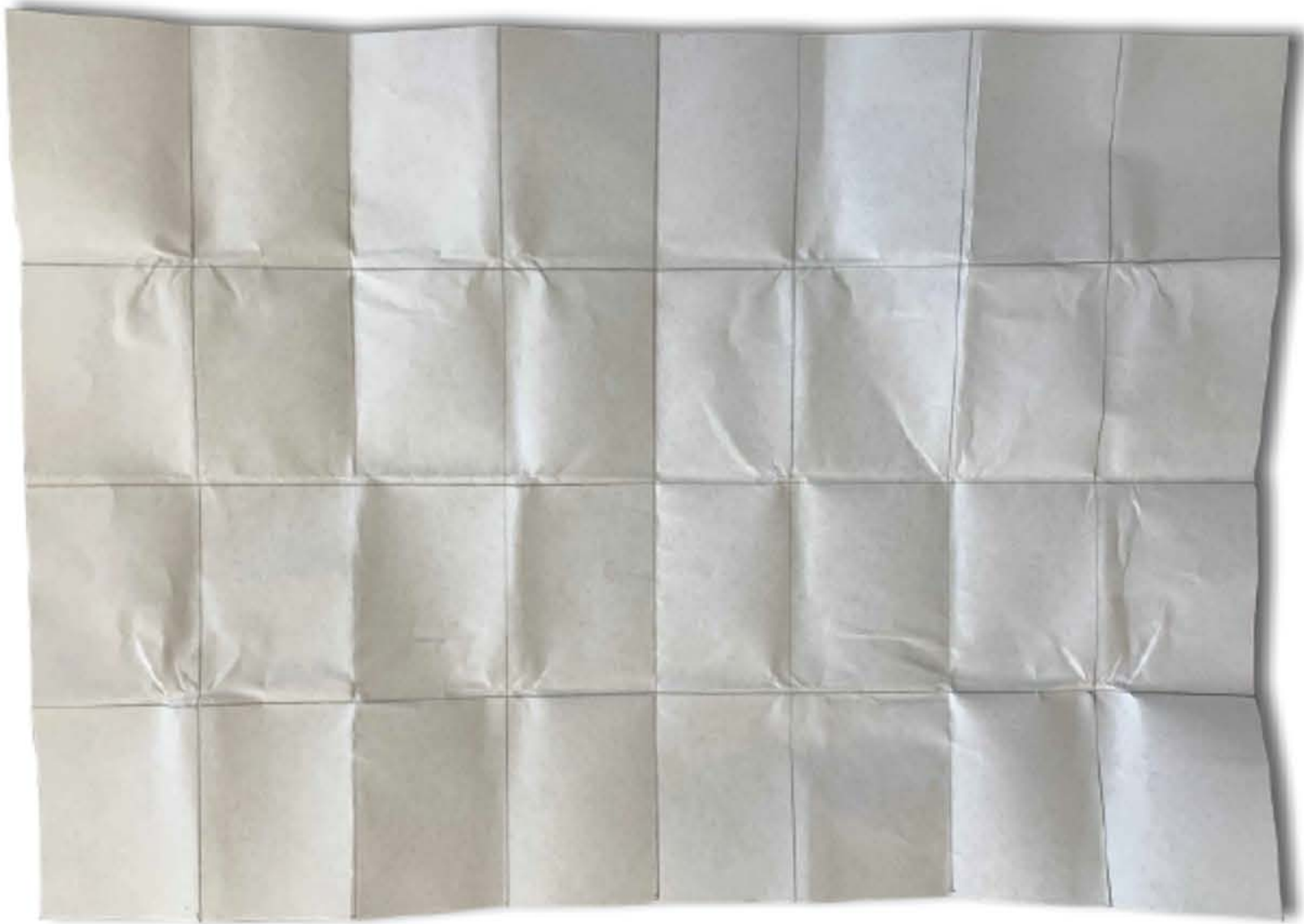


Max knew that vinegar is acidic and baking soda is alkaline. Max added some liquid he found in the kitchen to a new cup of cabbage juice. The cabbage juice turned red.

What can Max conclude from his experiment?

- ☐ The cabbage juice was acidic.
- ☐ The liquid contained vinegar.
- ☐ The liquid was acidic.
- ☐ The liquid was red.

Adam folded an unused piece of paper in half several times. The photograph shows what his paper looked like when he unfolded it.



How many times in total had he folded the paper?

☐

4

☐

5

☐

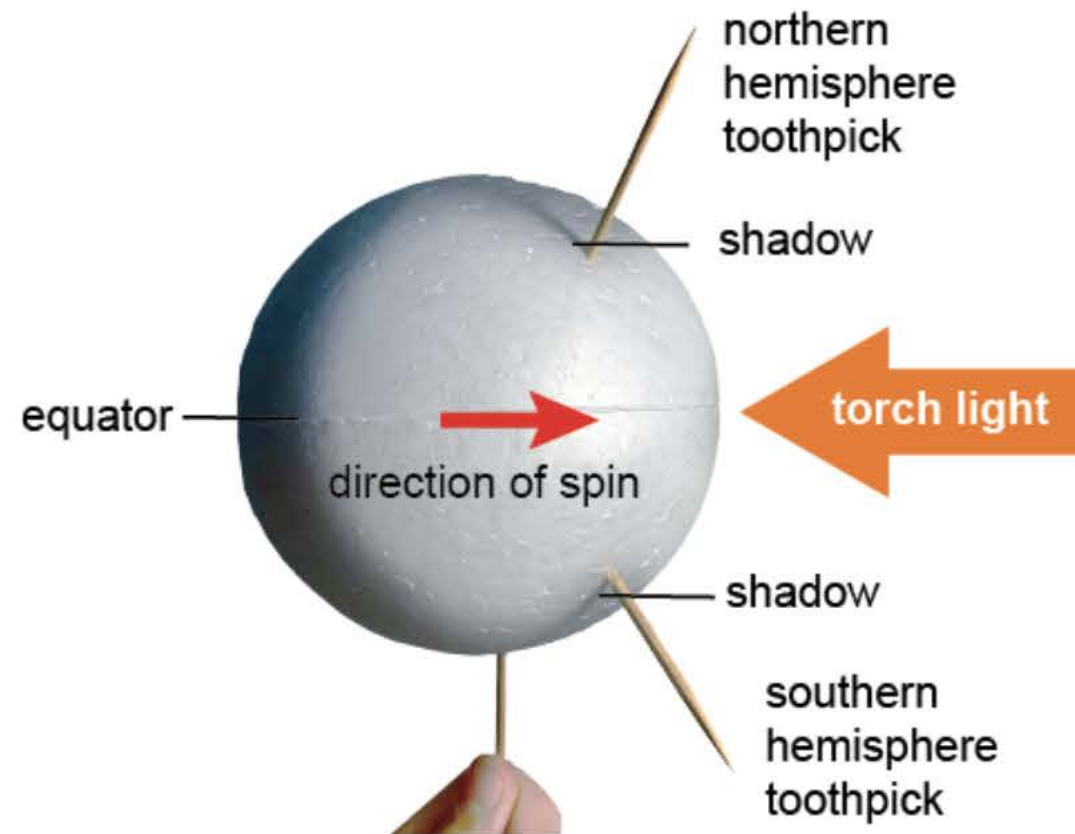
8

☐

16

Millie and Tom investigated shadows using a styrofoam ball on a stick to represent Earth.

They pushed one toothpick into the northern hemisphere and one toothpick into the southern hemisphere.



In a dark room, Tom shone a torch at the equator.

Millie twisted the stick to make the Earth spin in the direction shown.

The toothpick shadows moved either clockwise ↻ or anticlockwise ↻ around each toothpick.

Which of the following statements correctly describes the movement of the toothpicks' shadows?

- ☐ The northern hemisphere toothpick shadow moves clockwise and the southern hemisphere toothpick shadow moves clockwise.
- ☐ The northern hemisphere toothpick shadow moves clockwise and the southern hemisphere toothpick shadow moves anticlockwise.
- ☐ The northern hemisphere toothpick shadow moves anticlockwise and the southern hemisphere toothpick shadow moves anticlockwise.
- ☐ The northern hemisphere toothpick shadow moves anticlockwise and the southern hemisphere toothpick shadow moves clockwise.

Results

Question number	Correct answer	Mark	Category	Descriptor
1	A	☐ ☐	Observing & Measuring	Identify a photo of a space body by description
2	B	☐ ☐	Interpreting	Interpret a diagram of plant growth
3	D	☐ ☐	Observing & Measuring	Count the number of pairs of legs on a crab
4	A	☐ ☐	Reasoning & Problem solving	Use reasoning to infer the cause of a phenomenon
5	B	☐ ☐	Interpreting	Interpret a diagram illustrating an astronomical phenomenon
6	D	☐ ☐	Predicting & Concluding	Identify what affects the amount of energy captured by solar panels
7	C	☐ ☐	Observing & Measuring	Estimate half of a length given a full length
8	D	☐ ☐	Interpreting	Interpret a column graph about a student allergy survey
9	C	☐ ☐	Investigating	Identify the conditions for a fair test involving parachutes
10	B	☐ ☐	Predicting & Concluding	Predict the movement of the water level in response to an action
11	C	☐ ☐	Investigating	Identify three types of variables in a simple investigation
12	A	☐ ☐	Interpreting	Match the head of a screw to the correct bit
13	A	☐ ☐	Predicting & Concluding	Predict the position of a constellation based on information in diagrams
14	D	☐ ☐	Predicting & Concluding	Determine the criterion by which given animals have been grouped
15	B	☐ ☐	Observing & Measuring	Measure lengths using a ruler
16	A	☐ ☐	Observing & Measuring	Identify senses used in an observation
17	D	☐ ☐	Predicting & Concluding	Use a key to classify a caecilian
18	D	☐ ☐	Investigating	Identify whether a described investigation was a fair test
19	D	☐ ☐	Interpreting	Interpret results of an investigation into soil erosion
20	B	☐ ☐	Investigating	Identify the equipment needed for a valid investigation
21	B	☐ ☐	Reasoning & Problem solving	Deduce the direction of cogs
22	C	☐ ☐	Predicting & Concluding	Identify the correct set-up of a pooter to safely collect insects
23	D	☐ ☐	Interpreting	Interpret tabled information to label a diagram
24	B	☐ ☐	Reasoning & Problem solving	Determine the colour of shaded and unshaded areas under coloured lights
25	A	☐ ☐	Interpreting	Perform a calculation using data from a table
26	C	☐ ☐	Reasoning & Problem solving	Predict the shape of the shadow cast by a given object
27	C	☐ ☐	Investigating	Draw a conclusion based on the results of an investigation
28	B	☐ ☐	Reasoning & Problem solving	Determine the number of times a piece of paper was folded
29	B	☐ ☐	Reasoning & Problem solving	Solve a shadow problem by spatial reasoning

You have completed this practice test.

Your mark is

0 / 29

0%

Click here to reset the test and try again.



ICAS.